Bullying and Reading Achievement: A Comparison of South Africa and Brazil Using PIRLS 2021

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Abstract

This essay investigates the association between bullying frequency and reading achievement among Grade 4 learners in South Africa and Brazil, whether some learners are more strongly affected, and whether the South African pattern appears in another unequal upper-middle-income country. This is relevant in South Africa, where high reported bullying exposure coexists with low foundational literacy. Reading literacy is central to skill development and learners' future employment prospects. With the use of PIRLS 2021 data, survey-weighted linear regressions were estimated separately for both countries across five nested specifications that progressively control for learner, household and school characteristics. Furthermore, the interaction models examined differences by gender and home socioeconomic status. The results confirm the hypothesised negative association between bullying frequency and reading scores. In fully adjusted models, monthly bullying was associated with scores 36.1 points lower in South Africa and 38.4 points lower in Brazil than among learners categorised as "Never or almost Never bullied". Weekly bullying was associated with considerably larger gaps of 80.6 and 113.3 points, respectively. Adding controls reduced the estimated South African coefficients but produced little change in Brazil. Gender interactions were not statistically significant, whereas home SES interactions showed contrasting patterns. Higher SES intensified the monthly and weekly reading gaps associated with bullying in South Africa. In Brazil, the monthly interaction was insignificant, while higher SES reduced the weekly gap. Finally, common-sample and alternative-measure robustness checks produced similar findings in both countries. Despite the cross-sectional design limitation to permit causal conclusions, frequent bullying should be treated as an educational cause for concern.

1. Introduction

South Africa faces a severe early-grade reading crisis where 81% of Grade 4 learners tested in PIRLS 2021 could not read for meaning in any language (Spaull, 2023). Within the Schooling 2030 agenda, DBE-led initiatives have sought to improve early-grade literacy through structured pedagogy programmes combining lesson plans, teacher training and integrated reading materials (Department of Basic Education, nd). These instructional supports address one component of learners' educational environment. Another potentially important feature is learners'

exposure to bullying by their peers. According to the PIRLS 2021 results, approximately three quarters of South African learners reported experiencing bullying monthly or weekly (Mullis et al., 2023). The coexistence of poor reading achievement and widespread bullying raises an important educational and economic question about whether learners' exposure to bullying is associated with the development of their foundational reading skills. Existing evidence generally finds a negative relationship between exposure to bullying or school violence and educational achievement (Anton-Erxleben et al., 2016; Ozyildirim and Karadağ, 2024). Using PIRLS 2021 data, this essay examines the conditional association between bullying frequency and Grade 4 reading achievement in South Africa and Brazil. Brazil is used as a comparison country to assess whether the relationship observed in South Africa also appears in another unequal upper-middle-income economy with a comparable level of GDP per capita but substantially higher reading achievement (De Villiers and De Villiers, 2023; Spaull, 2023). The comparison therefore provides a benchmark for assessing whether the South African association is context-specific, although it does not hold all country-level differences constant. The central research question is: Is bullying frequency associated with Grade 4 reading achievement in South Africa, does this association differ by gender and home socioeconomic status, and how does the South African pattern compare with Brazil? Five nested regression specifications are estimated separately for each country, beginning with bullying frequency and progressively adding learner, household and school characteristics. Gender and home SES interactions are then used to examine whether the association differs across learners, while common-sample and alternative-measure specifications assess the robustness of the results.

2. Literature Review

A substantial body of international literature associates exposure to childhood violence with poorer educational outcomes. The systematic review and meta-analysis conducted by Fry et al. (2018) examined evidence from 21 countries and found that childhood violence was associated with lower academic achievement, greater absenteeism and an increased likelihood of school dropout.

More specific evidence on peer bullying is provided by Ozyildirim and Karadağ (2024), who conducted a meta-analysis using TIMSS and PIRLS results from 77 countries. Although the estimated associations were relatively small, bullying had its largest association with reading achievement when compared with mathematics and science. The association was also significantly larger among Grade 4 learners than among Grade 8 learners. These findings make the PIRLS Grade 4 sample particularly appropriate for examining the relationship between bullying frequency and reading achievement. Bullying is particularly prevalent in South African schools: approximately 73–74% of learners in PIRLS 2021 reported experiencing bullying about monthly or about weekly (Mullis et al., 2023). Historical evidence also indicates that bullying is associated with poorer educational achievement. Using PIRLS, prePIRLS and TIMSS 2011 data from Botswana, Ghana and South Africa, Anton-Erxleben et al. (2016) found substantial achievement differences between bullied and non-bullied learners after matching them on observed background characteristics.

Exposure to bullying is nevertheless not distributed evenly among South African learners. Using nationally representative TIMSS 2019 data, Juan et al. (2025) found that age, gender and

socioeconomic status were associated with the likelihood of experiencing different forms of victimisation. Boys and older learners generally faced a greater risk, whereas higher socioeconomic status provided some protection against most forms of bullying.

Brazil provides a useful comparison with South Africa because the two countries share several economic and educational characteristics. Approximately 42% of Brazilian learners reported experiencing bullying monthly or weekly in PIRLS 2021, considerably below the corresponding proportion in South Africa (Mullis et al., 2023). Regardless, both countries are upper-middle-income members of BRICS, that have highly unequal income distributions and have implemented comparable education policies (De Villiers and De Villiers, 2023; Spaull, 2023). Nevertheless, Brazil performs approximately three years ahead of South Africa in reading achievement. This contrast provides a basis for examining whether the association between bullying and reading performance differs between two unequal emerging economies with comparable policy contexts. Evidence from Brazil also supports the approach used in this essay. Using PISA 2015 data, Sbroglio Rizzotto and Aniceto França (2021) found that the relationship between bullying and school performance differed according to both the frequency and type of bullying experienced. Physical bullying and being threatened were associated with lower reading performance, while other forms showed weaker, insignificant or different results. These findings support the use of frequency categories in this essay, as learners who experience bullying weekly may be affected differently from those who experience it monthly or hardly ever.

Although Ireland is a substantially different context from South Africa, the longitudinal cohort study by Murphy et al. (2022) provides useful evidence on the relationship between bullying and achievement. The findings suggest that bullying at age nine predicted later psychological difficulties, which were associated with lower literacy and numeracy scores. This supports the proposed mechanism in which bullying affects achievement partly through learners' psychological difficulties. Psychological difficulties are therefore not included as a control in the main model, as doing so could remove part of the relationship this essay aims to estimate.

3. Data and Methodology

3.1 Data Source and Country Samples

This essay uses data from the 2021 Progress in International Reading Literacy Study (PIRLS), which assesses the reading achievement of Grade 4 learners. This study compares Brazil and South Africa as two upper-middle-income countries with similar levels of GDP per capita (Spaull, 2023). The analysis uses the South African and Brazilian student and school files from the public-use PIRLS 2021 International Database, Edition 2 (International Association for the Evaluation of Educational Achievement, 2024). The statistical analysis was conducted in `R` using reproducible scripts and the `EdSurvey` package.

3.2 Variables and Measurement

The outcome variable for this study is reading achievement, measured using all five plausible values (`ASRREA01`–`ASRREA05`). Plausible values are multiple draws from each learner's estimated achievement distribution and are intended to produce unbiased population-level esti-

mates rather than precise individual scores. The main explanatory variable is the bullying-frequency category (ASDGSB), which classifies learners as experiencing bullying "About Weekly," "About Monthly," or "Never or Almost Never." The continuous bullying scale (ASBGSB) is used as an alternative specification of the main explanatory variable. Higher scores indicate less frequent bullying.

Table 1: PIRLS 2021 data, analytical samples and variables

Item	PIRLS variable(s) or sample	Definition or use
Population and data	South African and Brazilian PIRLS 2021 student and school files	Grade 4 learners; one observation represents one learner
Analytical samples	M1: South Africa 11,660; Brazil 4,715. M5: South Africa 7,177; Brazil 3,563	M5 uses learners with complete information for all included controls; intermediate samples vary with data availability
Reading achievement	ASRREA01–ASRREA05	Five plausible values for overall reading achievement
Bullying	ASDGSB; ASBGSB	Main categories: never or almost never, monthly, or weekly; continuous scale used for robustness
Learner and household controls	ASBG01, ASDAGE, ASBHSES, ASBG03	Gender, age, home SES and frequency of speaking the assessment language at home
School controls	ACBG05B, ACBG03A	School location and principal-reported percentage of economically disadvantaged learners
Survey design	TOTWGT, JKZONE, JKREP	Student sampling weight and jackknife repeated-replication design

3.3 Empirical Strategy and Specifications

Five models were estimated using progressively larger sets of controls. Model 1 (**M1**) estimates the unadjusted association between bullying frequency and reading achievement. Model 2 (**M2**) adds controls for gender and age. Model 3 (**M3**) additionally controls for home socioeconomic status (SES), while Model 4 (**M4**) adds home language. Finally, Model 5 (**M5**) includes all the preceding variables as well as school location and school disadvantage.

The control variables were introduced in theoretically motivated blocks. Home resources were added as a measure of socioeconomic background, which is strongly associated with educational achievement and may also be related to bullying exposure (Sbroglio Rizzotto and Aniceto França, 2021). Home language was included because alignment between the language spoken at home, the language of instruction and the assessment language may influence reading performance and may also capture relevant cultural or minority-group differences. Finally, school location and school disadvantage were included to account for differences in school resources, socioeconomic composition and safety environments that may be associated with both bullying and achievement (Anton-Erxleben et al., 2016; Juan et al., 2025).

Because the complete-case requirements become more restrictive as controls are added, the analytical sample decreases in Models 3 and 5. As a robustness check, Models 1–4 were therefore re-estimated using the complete-case sample from Model 5. Holding the sample constant helps distinguish changes in the estimated bullying coefficients resulting from covariant adjustment from changes caused by differences in sample composition.

The models are estimated separately for South Africa and Brazil. For country c, plausible value p, and specification m, the estimating equation is

$$Y_{isc}^{(p)} = \alpha_c^{(m)} + \beta_{1c}^{(m)}\text{Monthly}_{isc} + \beta_{2c}^{(m)}\text{Weekly}_{isc} + \mathbf{X}'_{isc,m}\boldsymbol{\gamma}_c^{(m)} + \varepsilon_{isc}^{(p)}, \qquad p = 1, \ldots, 5, \qquad (1)$$

In Equation 1, $Y_{isc}^{(p)}$ denotes reading plausible value p for learner i attending school s in country c. The indicators Monthly_{isc} and Weekly_{isc} identify learners classified as experiencing bullying about monthly or about weekly, respectively, with learners who were never or almost never bullied forming the omitted reference group. The corresponding coefficients, $\beta_{1c}^{(m)}$ and $\beta_{2c}^{(m)}$, therefore measure conditional mean differences in PIRLS reading points relative to this reference group. The vector $\mathbf{X}_{isc,m}$ contains the controls included in specification m, while $\boldsymbol{\gamma}_c^{(m)}$ contains their associated coefficients. Two additional specifications interact bullying frequency with gender (I1) and home socioeconomic status (I2) to assess whether the bullying–achievement association differs across these groups.

3.4 Survey Design and Estimation

All estimates incorporate the PIRLS total student sampling weight (TOTWGT), which accounts for unequal probabilities of selection and adjustments for nonresponse at the school, class and student levels. Sampling variances are estimated using the PIRLS jackknife repeated replication procedure and the supplied JKZONE and JKREP variables. This procedure accounts for the clustered and stratified survey design. Reading-achievement estimates additionally incorporate uncertainty across the five plausible values.

3.5 Methodological Limitations

Several limitations should be considered. First, bullying exposure and reading achievement were measured during the same assessment period. The analysis therefore cannot determine which occurred first. Bullying may contribute to poorer achievement, but lower-achieving learners may also face a greater risk of victimisation. Unobserved family, peer and school characteristics may additionally influence both bullying and achievement (Anton-Erxleben et al., 2016).

Second, bullying exposure is self-reported and may therefore be affected by recall error, differences in learners' interpretation of the questions and their willingness to report victimisation.

Third, excluding learners with missing information may introduce selection bias if missingness is related to bullying or reading achievement. The smaller sample may also result in less precise estimates.

Fourth, South African and Brazilian Grade 4 learners were assessed one year later than originally scheduled, during a period of widespread COVID-19-related educational disruption.

Finally, Brazil and South Africa should not be treated as interchangeable simply because both are classified as upper-middle-income countries. They differ in their education systems, levels and patterns of inequality, linguistic environments, school conditions and social understandings of bullying. The estimated associations are therefore context-specific averages and may conceal substantial heterogeneity across learners and schools. Evidence from these two countries cannot be generalised to all upper-middle-income countries.

4. Results

Table 2 presents the M1–M5 estimates for South Africa and Brazil. The full set of control coefficients is reported in Appendix Table 4. South Africa in Model 1 that just controls for Bullying Penalty indicates a decrease of 55.1 points in reading scores if the learner is bullied on a monthly basis and a decrease of 116.6 points if bullied on a weekly basis. Brazil displays the same negative pattern: learners bullied monthly score 36.4 points lower, while those bullied weekly score 123.5 points lower. All four M1 bullying coefficients are statistically significant at the 1% level.

The model develops in interesting directions as controls are added in M2, M3, M4 and M5. In South Africa, the estimated bullying penalties become progressively smaller as controls are added, with the M5 coefficients indicating a 36.1-point decrease for monthly bullying and an 80.6-point decrease for weekly bullying. Brazil, by contrast, remains comparatively stable: the monthly penalty increases in magnitude from 36.4 points in M1 to 38.4 points in M5, while the weekly penalty decreases from 123.5 to 113.3 points. The standard errors generally decrease as controls are introduced, indicating greater precision. The explained variance increases from 13.4% to 38.0% in South Africa and from 13.9% to 35.5% in Brazil.

Table 2: Bullying coefficients across the main model sequence

	South Africa					Brazil				
	M1	M2	M3	M4	M5	M1	M2	M3	M4	M5
Bullied about monthly	-55.1***	-53.8***	-46.8***	-43.9***	-36.1***	-36.4***	-37.5***	-39.3***	-36.5***	-38.4***
	(6.9)	(6.8)	(5.9)	(5.5)	(5.5)	(5.7)	(5.4)	(5.5)	(5.3)	(5.8)
Bullied about weekly	-116.5***	-110.4***	-97.9***	-94.2***	-80.6***	-123.5***	-119.2***	-112.9***	-109.4***	-113.3***
	(7.8)	(7.5)	(6.8)	(6.7)	(6.7)	(7.6)	(8.0)	(7.3)	(7.2)	(7.9)
Learners	11,660	11,618	9,167	8,505	7,177	4,715	4,706	4,110	4,022	3,563
R^2	0.134	0.170	0.278	0.296	0.380	0.139	0.165	0.317	0.324	0.355

Notes: Entries are coefficients with jackknife standard errors in parentheses. The reference group is almost never bullied. All models use the default PIRLS student weight and all five reading plausible values. M1 includes bullying; M2 adds gender and age; M3 adds continuous home SES; M4 adds home language; M5 adds school location and economically disadvantaged composition. Full coefficients are reported in Appendix Table 4. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Appendix Figure 1 summarizes the bullying coefficients across M1–M5, showing how the estimated associations change as successive controls are introduced.

Re-estimating M1–M4 on the common M5 sample produced a similar pattern of coefficients, suggesting that the changes across models were not primarily attributable to sample composition (Appendix Table 5). The continuous-scale specification also produced results consistent with the main analysis (Appendix Table 6).

Table 3 presents the interactions between bullying and gender (I1) and between bullying and home SES (I2). The gender interactions are not statistically significant in either South Africa or Brazil. There is therefore no evidence that the association between bullying and reading achievement differs between boys and girls.

The SES interaction produces a different pattern. In South Africa, each one-unit increase in home SES makes the estimated monthly bullying penalty 13.2 points more negative and the weekly penalty 20.3 points more negative. In Brazil, the corresponding interaction coefficients are positive: 3.2 points for monthly bullying and 9.0 points for weekly bullying. The monthly interaction is not statistically significant, whereas the weekly interaction is significant at the 5%

level. Thus, there is evidence that higher SES intensifies the bullying penalty in South Africa, while the Brazilian estimates provide limited evidence that higher SES moderates the weekly bullying penalty.

Table 3: Heterogeneity in the association between bullying and reading achievement

	South Africa		Brazil	
	I1	I2	I1	I2
Bullied about monthly	-46.5***	-42.2***	-34.1***	-39.8***
	(7.0)	(5.7)	(8.8)	(5.6)
Bullied about weekly	-95.3***	-95.5***	-120.4***	-110.9***
	(7.8)	(6.4)	(10.6)	(6.9)
Boy	-41.2***	-42.9***	-4.1	-5.6
	(6.7)	(3.4)	(4.9)	(4.7)
Age (years)	-7.6**	-8.0**	-12.9***	-13.5***
	(3.4)	(3.4)	(4.4)	(4.3)
Home SES (ASBHSES scale points)	25.7***		26.2***	
	(2.3)		(2.1)	
Bullied about monthly × Boy	-0.7		-10.5	
	(9.4)		(11.1)	
Bullied about weekly × Boy	-5.1		9.9	
	(7.4)		(12.7)	
Home SES (centred scale points)		37.5***		24.1***
		(2.8)		(2.0)
Bullied about monthly × Home SES (centred scale points)		-13.2***		3.2
		(2.8)		(3.0)
Bullied about weekly × Home SES (centred scale points)		-20.3***		9.0**
		(2.8)		(4.5)
Other gender category				-86.7**
				(35.2)
Learners	9,167	9,167	4,076	4,110
R^2	0.278	0.289	0.314	0.319

Notes: Entries are coefficients with jackknife standard errors in parentheses. I1 interacts bullying with gender and excludes Brazil's 41 learners in the Other gender category. I2 interacts bullying with continuous ASBHSES centred at the design-weighted country mean; one unit is one PIRLS ASBHSES scale point. The reference bullying category is almost never and the gender reference is girls. Significance: *** p<0.01, ** p<0.05, * p<0.10.

5. Discussion

The results demonstrate a strong negative association between bullying frequency and Grade 4 reading achievement in both South Africa and Brazil. These associations remained statistically significant after controlling for learner, household and school characteristics. The findings therefore support the main hypothesis and are consistent with previous international and African evidence (Fry et al., 2018; Ozyildirim and Karadağ, 2024; Anton-Erxleben et al., 2016).

The coefficients followed different patterns as controls were introduced. In South Africa, the estimated bullying penalties became progressively smaller, whereas the Brazilian estimates remained comparatively stable. This suggests that observed learner, household and school characteristics account statistically for part of the South African association but explain less of the Brazilian association. It does not establish that the remaining association is causal, because unobserved differences may still influence both bullying and achievement. Nevertheless, the contrast cautions against treating upper-middle-income countries as educationally homogeneous.

Several mechanisms may help explain the negative association. Previous research also identifies absenteeism, reduced school engagement, poor concentration and feelings of insecurity as possible pathways linking bullying to achievement (Fry et al., 2018; Anton-Erxleben et al., 2016). These mechanisms could cause bullied learners to fall behind academically, but the present analysis does not test which pathway is primarily responsible.

Although previous South African research finds that boys are generally more likely to experience bullying than girls (Juan et al., 2025), the gender interactions in the present analysis were not statistically significant. These findings are not contradictory: gender may shape learners' exposure to particular forms of bullying without changing the association between a given frequency of bullying and reading achievement.

In Brazil, higher SES appeared to reduce the weekly bullying penalty, although the monthly interaction was not significant. In South Africa, the monthly and weekly penalties became larger as SES increased. One possible explanation is that bullied high-SES learners fall further behind the comparatively high scores of non-bullied learners from similar backgrounds. This explanation remains tentative because the analysis cannot identify the mechanisms producing the interaction.

Taken together, the findings indicate that bullying should be treated as an educational concern as well as a learner well-being concern. This is consistent with the Sustainable Development Goal commitments to end violence against children and promote safe, non-violent and inclusive learning environments (Fry et al., 2018). Because the data are cross-sectional, the estimates should not be interpreted causally, and unobserved characteristics may influence both bullying exposure and reading achievement. Nevertheless, the common-sample and continuous-scale robustness checks produced similar patterns, reducing concerns that the main findings are solely artefacts of changing samples or the categorical bullying measure.

6. Conclusion

The estimated coefficient revealed statistically significant and large conditional reading score gaps for learner that experienced bullying Monthly and Weekly, with more severe gaps among learners with Weekly bullying. South African coefficients experienced attenuation as the learner, school and household characteristics were controlled reflecting the observable differences between learners. The Brazilian coefficients remained stable suggesting that the included covariates account for less of the association in the sample. The interaction models revealed that, although gender may influence exposure to bullying, it did not significantly change the association between bullying frequency and reading scores. However, the country-specific home SES interactions showed contrasting patterns: higher SES intensified the monthly and weekly bullying penalties in South Africa but attenuated the weekly penalty in Brazil, while the Brazilian monthly interaction was not statistically significant. Nevertheless, the nature of cross section analysis does not address reverse causality, nor omitted variable bias. Subject to these limitations, the evidence suggests that bullying is associated with lower reading achievement and should be recognised as an important educational concern.

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Appendix

Table 4: Full bullying and reading achievement model sequence

	South Africa					Brazil				
	M1	M2	M3	M4	M5	M1	M2	M3	M4	M5
Bullied about monthly	-55.1***	-53.8***	-46.8***	-43.9***	-36.1***	-36.4***	-37.5***	-39.3***	-36.5***	-38.4***
	(6.9)	(6.8)	(5.9)	(5.5)	(5.5)	(5.7)	(5.4)	(5.5)	(5.3)	(5.8)
Bullied about weekly	-116.5***	-110.4***	-97.9***	-94.2***	-80.6***	-123.5***	-119.2***	-112.9***	-109.4***	-113.3***
	(7.8)	(7.5)	(6.8)	(6.7)	(6.7)	(7.6)	(8.0)	(7.3)	(7.2)	(7.9)
Boy		-44.4***	-43.4***	-40.8***	-43.1***		-10.2**	-5.4	-3.9	-4.9
		(3.1)	(3.5)	(3.8)	(3.9)		(5.0)	(4.6)	(4.5)	(4.8)
Age (years)		-10.0***	-7.6**	-6.9**	-7.6**		-24.6***	-13.1***	-12.6***	-12.7**
		(3.0)	(3.4)	(3.4)	(3.3)		(3.9)	(4.4)	(4.6)	(4.8)
Home SES (ASBHSES scale points)			25.7***	24.9***	16.8***			26.3***	25.7***	21.6***
			(2.3)	(2.4)	(2.5)			(2.0)	(2.0)	(1.8)
Home language: almost always test language				7.4	-5.5				-18.7*	-20.8*
				(8.1)	(8.1)				(10.7)	(10.7)
Home language: sometimes test language				46.3***	22.4***				7.6	12.3
				(7.4)	(7.1)				(9.8)	(9.5)
Home language: never test language				-28.0***	-40.7***				-93.2***	-83.3***
				(8.9)	(10.6)				(19.2)	(19.4)
School location: suburban					-18.0					-1.3
					(17.6)					(15.8)
School location: medium city/large town					1.9					-18.6**
					(34.0)					(9.1)
School location: small town/village					-45.1**					-25.3**
					(18.8)					(11.5)
School location: remote rural					-55.5***					-14.4
					(16.5)					(25.3)
Economically disadvantaged: 11 to 25%					-49.4*					-25.5**
					(27.8)					(10.9)
Economically disadvantaged: 26 to 50%					-82.8***					-47.6***
					(20.2)					(13.7)
Economically disadvantaged: more than 50%					-117.3***					-40.9***
					(17.1)					(11.7)
Other gender category							-100.2***	-86.7**	-85.6**	-49.1**
							(31.1)	(35.1)	(35.6)	(23.6)
Learners	11,660	11,618	9,167	8,505	7,177	4,715	4,706	4,110	4,022	3,563
R^2	0.134	0.170	0.278	0.296	0.380	0.139	0.165	0.317	0.324	0.355

Notes: Entries are coefficients with jackknife standard errors in parentheses. The reference group is almost never bullied; the gender reference is girls. All models use the default PIRLS student weight and all five reading plausible values. M1 includes bullying; M2 adds gender and age; M3 adds continuous home SES; M4 adds home language; M5 adds school location and economically disadvantaged composition. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 5: Common-sample sensitivity analysis

	South Africa				Brazil			
	M1	M2	M3	M4	M1	M2	M3	M4
Bullied about monthly	-51.5***	-50.7***	-41.9***	-41.2***	-37.8***	-38.6***	-39.4***	-37.9***
	(8.6)	(8.5)	(6.8)	(6.5)	(6.8)	(6.6)	(6.0)	(5.8)
Bullied about weekly	-112.5***	-107.8***	-91.7***	-89.9***	-131.2***	-128.2***	-116.8***	-114.6***
	(9.7)	(9.5)	(7.8)	(7.5)	(8.5)	(8.7)	(8.0)	(7.7)
Boy		-45.5***	-43.1***	-41.4***		-5.9	-4.5	-3.2
		(4.3)	(4.1)	(4.1)		(5.7)	(4.9)	(4.8)
Age (years)		-5.9	-5.5	-6.6*		-24.7***	-13.8***	-13.5***
		(3.9)	(3.7)	(3.6)		(5.2)	(4.7)	(4.7)
Home SES (ASBHSES scale points)			26.5***	25.4***			25.7***	25.7***
			(2.6)	(2.6)			(2.2)	(2.2)
Home language: almost always test language				9.1				-20.2*
				(9.2)				(11.4)
Home language: sometimes test language				46.8***				12.4
				(7.9)				(9.6)
Home language: never test language				-24.6**				-87.7***
				(10.1)				(19.4)
Other gender category						-72.9***	-53.4**	-50.2**
						(24.3)	(24.0)	(24.4)
Learners	7,177	7,177	7,177	7,177	3,563	3,563	3,563	3,563
R^2	0.123	0.156	0.275	0.296	0.156	0.177	0.317	0.328

Notes: Entries are coefficients with jackknife standard errors in parentheses. All models use the M5 complete-case sample. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Table 6: Continuous-bullying robustness model

	South Africa	Brazil
	R1	R1
Bullying scale (higher = less bullying)	18.7***	16.8***
	(1.3)	(1.2)
Boy	-43.7***	-5.0
	(3.6)	(4.8)
Age (years)	-8.0**	-13.7***
	(3.4)	(4.9)
Home SES (ASBHSES scale points)	26.1***	27.4***
	(2.3)	(2.1)
Other gender category		-97.7**
		(36.8)
Learners	9,167	4,110
R^2	0.270	0.293

Notes: Entries are coefficients with jackknife standard errors in parentheses. Higher ASBGSB values indicate less bullying. Significance: *** p<0.01, ** p<0.05, * p<0.10.

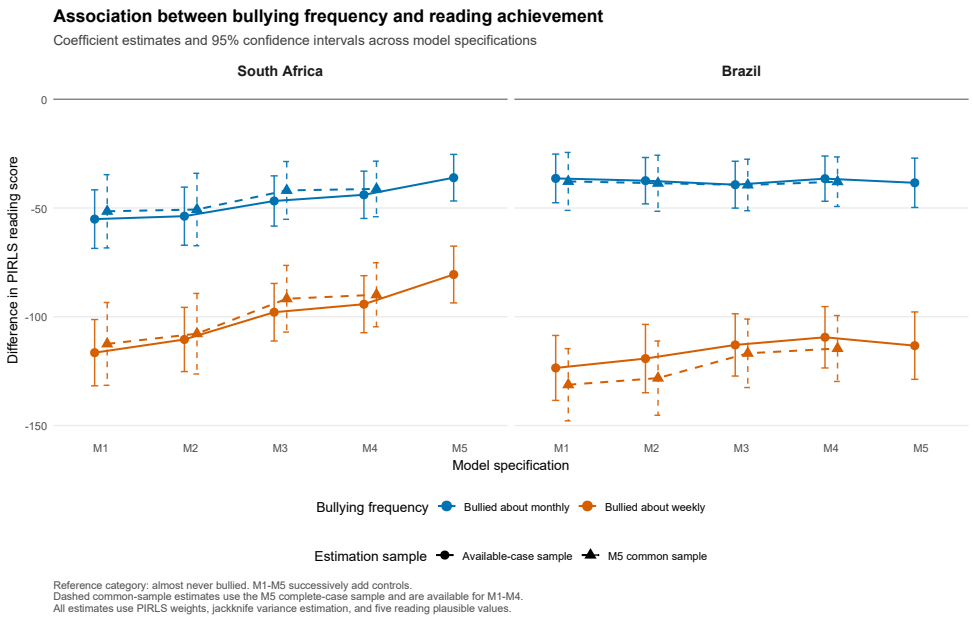


Figure 1: Estimated association between bullying frequency and reading achievement across model specifications